Haoyue Tang

Employment

University of Hawaii at Manoa, Honolulu, HI, USA

Assistant Professor, Department of Electrical and Computer Engineering, 2026.08-Now

Meta Platforms, Inc., Menlo Park, CA, USA

Senior Research Scientist, Ranking Foundational AI (CoreML), 2024.07-2026.08

Apply SOTA ML techniques to Ads Ranking

- User modeling using hierarchical transformers, software-hardware co-design to improve key CVR prediction metrics in head models, achieving 0.1% Global Ads Score Gain (GAS).

- Continual learning and distillation to improve model generalization, improve key CVR predict metrics in head models by 0.05% GAS gain.

Technical University Munich and Technical University Berlin, German

Visiting Scholar, 2024.06

Sponsored by DAAD-NetAI Fellowship

Meta AI (Formerly Facebook Applied AI Research), Sunnyvale, CA, USA

Postdoctoral Associate, 2023.04-2024.04

Diffusion Models and Reinforcement Learning for Improving the Image Super-Resolution, Deblurring and Inpainting Quality (AISTATS2024, NeurIPS2023 Workshop)

Yale Univeresity, **NSF AI Institute for Edge Computing Leveraging Next Generation Networks**

Postdoctoral Research Associate, 2022.01-2023.04

Online Learning for Data Freshness Optimization in Networking Systems (Mobihoc 2022, IEEE Trans on Information Theory, IEEE Trans on Networking)

Microsoft Research, Beijing, China

Research Intern, "Stars of Tomorrow", 2020.11-2021.11

Improve out-of-distribution and transfer learning accuracy using causal learning (NeurIPS 2021)

Develop new dual generation models using joint probability compatibility (NeurIPS 2021)

Education

Tsinghua University, Beijing, China

Ph.D.(with honors), Department of Electrical Engineering, 2017 to 2021

B.S., Electrical Engineering, 2013-2017

Research on wireless networking, with emphasis on data freshness oriented scheduling, online learning and mechanism design.

Grant and Project Funding

PI, NSF ACCESS Allocations Program, Explore Allocation 2026–2027

– ELE260074: Diagnosing and Improving Conditional Score Estimation for Classifier-Free Guidance

– ACCESS Credits: 200,000

PI, NSF NAIRR Pilot, Classroom Allocation 2026–2027

– NAIRR260411: GPU Resources for Teaching Information-Theoretic Deep Generative Models (ECE 693B)

– Indiana Jetstream2 GPU: 200,000 SUs

PI, NSF NAIRR Pilot, Start-Up Allocation 2026–2027

– NAIRR260485: Start-Up Allocation, Artificial Intelligence and Intelligent Systems

– TACC Vista (NVIDIA GH100 Grace Hopper Superchip): 2,000 Node Hours

Publication

Refereed Journal Articles

1. H. He, **H. Tang**, J. Pan, J. Wang, J. Song and L. Tassiulas, "Age Optimal Sampling for Unreliable Channels Under Unknown Channel Statistics," in *IEEE Transactions on Mobile Computing*, vol. 25, no. 6, pp. 7640-7656, June 2026.

2. **H. Tang**, Y. Sun, and L. Tassiulas, "Sampling of the Wiener Process for Remote Estimation over a Channel with Unknown Delay Statistics", in *IEEE/ACM Transactions on Networking*, vol. 32, no. 03, pp. 1920-1935, 2024.

3. Y. Chen, **H. Tang**, J. Wang, P. Yang and L. Tassiulas, "Sampling for Remote Estimation of an Ornstein-Uhlenbeck Process through Channel with Unknown Delay Statistics", in *Journal of Communications and Networks*, vol. 25, no. 5, pp. 670-687, Oct. 2023.

4. S. Yang, L. Li, **H. Tang** and J. Wang, "Tradeoff between Diversity and Multiplexing Gains in Block Fading Optical Wireless Channels," in *IEEE Transactions on Information Theory*, vol. 69, no. 6, pp. 3415-3430, June 2023.

5. **H. Tang**, Y. Chen, J. Wang, P. Yang and L. Tassiulas, "Age Optimal Sampling Under Unknown Delay Statistics," in *IEEE Transactions on Information Theory*, vol. 69, no. 2, pp. 1295-1314, Feb. 2023.

6. Y. Chen, **H. Tang**, J. Wang, and J. Song, "Optimizing Age Penalty in Time-Varying Networks with Markovian and Error-Prone Channel State," *Entropy*, vol. 23, no. 1, p. 91, Jan. 2021.

7. **H. Tang**, J. Wang, L. Song and J. Song, "Minimizing Age of Information With Power Constraints: Multi-User Opportunistic Scheduling in Multi-State Time-Varying Channels," in *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 5, pp. 854-868, May 2020.

8. **H. Tang**, J. Wang, Z. Tang and J. Song, "Scheduling to Minimize Age of Synchronization in Wireless Broadcast Networks With Random Updates," in *IEEE Transactions on Wireless Communications*, vol. 19, no. 6, pp. 4023-4037, June 2020.

9. **H. Tang**, J. Wang and L. He, "Off-Grid Sparse Bayesian Learning-Based Channel Estimation for MmWave Massive MIMO Uplink," in *IEEE Wireless Communications Letters*, vol. 8, no. 1, pp. 45-48, Feb. 2019.

Refereed Conference Articles

10. M. Pan, **H. Tang**, J. Pan, T. Qiu and J. Yin, "Regret-Optimal and Stability-Enhanced Online Sampling of the Wiener Process for Remote Estimation over an Unreliable Channel with Unknown Statistics," *IEEE INFOCOM 2026 - IEEE Conference on Computer Communications, Tokyo, Japan, 2026, pp. 1-10*

11. **H. Tang**, T. Xie, A. Feng, H. Wang, C. Zhang and Y. Bai, "Solving General Inverse Problems via Posterior Sampling: A Policy Gradient Viewpoint", AISTATS 2024.
12. C. Zheng, **H. Tang**, M. Zang, X. Hong. A. Feng, L. Tassiulas and N. Zilberman, "DINC: toward distributed in-network computing", CoNEXT 2023.
13. **H. Tang**, T. Xie, A. Feng, H. Wang, C. Zhang and Y. Bai, "Solving Inverse Problems via Posterior Sampling: A Policy Gradient Viewpoint", accepted 2023 NeurIPS workshop on Deep Learning for Differential Equaitons (DLDE)
14. H. He, **H. Tang**, J. Pan, J. Wang, J. Song and L. Tassiulas, "Age optimal sampling for unreliable channels under unknown channel statistics", WiOPT2023 workshop on Semantic Communications.
15. **H. Tang**, Y. Sun, and L. Tassiulas, "Sampling of the Wiener Process for Remote Estimation over a Channel with Unknown Delay Statistics", In *the Twenty-second International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc '22)*, October 17–20, 2022, Seoul, Republic of Korea. ACM, New York, NY, USA. (Acceptance rate: 24/160=15%)
16. **H. Tang**, Y. Chen, J. Wang, J. Sun and J. Song, "Sending Timely Status Updates through Channel with Random Delay via Online Learning", *IEEE INFOCOM 2022 - IEEE Conference on Computer Communications*, 2022, pp. 1819-1827. (acceptance rate 19.9%)
17. X. Hou, **H. Tang**, J. Wang and J. Song, "Information Directed Learning Algorithm for Minimizing Queue Length Regret," *2022 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB)*, 2022, pp. 1-5.
18. Y. Chen, **H. Tang**, J. Wang and J. Song, "Adaptive Link Rate Selection for Throughput Maximization with Batched Thompson Sampling," *2022 IEEE International Conference on Communications Workshops (ICC Workshops)*, 2022, pp. 1-6.
19. **H. Tang**, X. Hou, J. Wang and J. Song, "Joint Link Rate Selection and Channel State Change Detection in Block-Fading Channels", *2021 IEEE Global Communications Conference (GLOBECOM)*, 2021, pp. 1-6.
20. C. Liu, **H. Tang**, T. Qin, J. Wang, Tie-Yan Liu, " On the Generative Utility of Cyclic Conditionals, " Advances in Neural Information Processing Systems 34 (2021): 6155-6170.
21. C. Liu, X. Sun, J. Wang, **H. Tang**, T. Li, T. Qin, W. Chen, and T. Liu, "Learning Causal Semantic Representation for Out-of-Distribution Prediction", Advances in Neural Information Processing Systems 34 (2021): 6155-6170.
22. **H. Tang**, P. Ciblat, J. Wang, M. Wigger and R. Yates, "Cache Updating Strategy Minimizing the Age of Information with Time-Varying Files' Popularities. ," *2020 IEEE Information Theory Workshop (ITW)*, 2021, pp. 1-5.
23. Y. Tan, **H. Tang**, J. Wang and J. Song, "Early Drop: A Packet-Dropping Incentive Rate Control Mechanism to Keep Data Fresh under Heterogeneous QoS Requirements, " *2021 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB)*, 2021, pp. 1-6.
24. **H. Tang**, P. Ciblat, J. Wang, M. Wigger and R. Yates, "Age of Information Aware Cache Updating with File- and Age-Dependent Update Durations," *2020 18th International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOPT)*, Volos, Greece, 2020, pp. 1-6.
25. Y. Chen, **H. Tang**, J. Wang, "Optimizing Age of Information in Multicast Unilateral Networks", *2020 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB)*, Virtual, 2020, pp. 1-4.
26. W. Zhan, **H. Tang**, J. Wang, "Delay Optimal Cross-Layer Scheduling Over Markov Channels with Power Constraint", *2020 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB)*, Virtual, 2020, pp. 1-4.
27. Q. Zhang, **H. Tang** and J. Wang, "Minimizing the Age of Synchronization in Power-Constrained Wireless Networks with Unreliable Time-Varying Channels," *IEEE INFOCOM 2020 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, Toronto, ON, Canada, 2020, pp. 936-941.

28. **H. Tang**, J. Wang, L. Song and J. Song, "Scheduling to Minimize Age of Information in Multi-State Time-Varying Networks with Power Constraints," *2019 57th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, Monticello, IL, USA, 2019.
29. **H. Tang**, J. Wang, Z. Tang and J. Song, "Scheduling to Minimize Age of Synchronization in Wireless Broadcast Networks with Random Updates," *2019 IEEE International Symposium on Information Theory (ISIT)*, Paris, France, 2019, pp. 1027-1031.
30. **H. Tang**, J. Wang and Y. R. Zheng, "Covert communications with extremely low power under finite block length over slow fading," *IEEE INFOCOM 2018 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, Honolulu, HI, 2018, pp. 657-661.
31. **H. Tang**, J. Wang and L. He, "Mutual information maximization for optimal spatial modulation MIMO system," *2017 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB)*, Cagliari, 2017, pp. 1-4.

Awards

Apr. 2023	DAAD Postdoc-NeT-AI Fellowship
Oct. 2022	EECS Rising Star, UT Austin
Apr. 2022	North American School of Information Theory Travel Grant
Apr. 2022	Zijing Scholar
Mar. 2022	Infocom 2022 Student Grant
Jan. 2022	Excellent Graduate of Beijing
Dec. 2021	Women in Communications Engineering (WICE) Travel Grant for Globecom2021
Dec. 2021	Microsoft Research Asia "Stars of Tomorrow" Excellent Intern
Apr. 2021	Tencent "Wizz Bird" Elite Research Student Program
Dec. 2020	Stars of Haiying (awarded to PhD candidates in Haidian District, top 1 in each department)
Oct. 2020	National Scholarship (top 2% in EE Department)
Oct. 2020	Undergraduate Student Research Training First Prize (to my students Q. Zhang and W. Zhan)
Nov. 2019	Best Paper Award, Department of Electronic Engineering PhD Forum, Tsinghua University
Sep. 2019	First Class Scholarship for Comprehensive Excellence, Tsinghua University
Jul. 2019	"Deng-Feng" Travel Award, Tsinghua University (for ISIT2019)
Apr. 2019	Stars of EE Department (Highest Honor for Graduate Student in EE Department for Research Excellence and Specialities), Tsinghua University
2014-2016	Zhou Huiqi Scholarship for overall excellence, Tsinghua University
May. 2015	China Scholarship Council Excellent **Undergraduate** Fellowship
Nov. 2012	Silver Medal in 29th National Physics Olymplics Contest(Ranking 61 out of 92000)

Invited Talk

Semantic Networking: Sensing, Scheduling and Signal Processing for Fresh Data

Seminar, School of Information Science and Engineering, Southeast University, July 2024.

Seminar, School of Computation, Information and Technology, Technical University of Munich, June 2024.

Seminar, Department of Electrical and Computer Engineering, University of Hawaii at Manoa, May 2024.

Seminar, Department of Computer Science, City University of New York, April 2024.
Seminar, Department of Engineering Technology, University of Houston, Feb. 2024.
Seminar, Department of ECE, Missouri University of Science and Technology, Feb. 2024.

Solving Inverse Problems with Pretrained Diffusion Models: A Policy Gradient Viewpoint
Department of Electrical Engineering and Yale Institute for Networking Research, Yale University, April 2023.
Seminar, Meta AI, Aug. 2023

Automated Networks Configurations for Time-Sensitive Applications
Seminar, Notre Dame University, Feb. 2023

Data Freshness Oriented Communication System Design: From Sampling to Scheduling
Seminar, Auburn University, Dec. 2022

Online Sampling for Data Freshness Optimization
EECS Rising Stars Workshop, UT Austin, Austin, Oct. 2022
Tsinghua University EE Department Graduate Forum/ Tsinghua-Berkeley Shenzhen Institute, May 2022
Graduation Day, Information Theory and Applications Workshop (ITA), UCSD, May 2022
Seminar, Institute of Network Science, Yale University, Jan 2022

Data Freshness oriented Scheduling in Time-Varying Networks
Young Researcher Forum, Sun Yat-Sen University, Dec. 2019
Seminar, School of Information Science, Shanghai University of Science and Technology, Oct. 2021

Teaching Experience

ECE693B: Information Theoretic Methods in Generative AI Fall 2026
Instructor, Department of Electrical and Computer Engineering, University of Hawaii at Manoa

Communication Signal Processing Spring 2021
Teaching Assistant, with Prof. Jintao Wang, at Tsinghua University

Communication Networks Fall 2020
Guest Lecture on Information Freshness, with Prof. Jintao Wang, at Tsinghua University

Former Undergraduate Students working on Projects with me
Yuchao Chen: PhD student@Tsinghua University.
Chenghao Deng: PhD student@University of Maryland, College Park –> Machine Learning Engineer@TikTok.
Wenhao Zhan: PhD student@Princeton –> Databricks Inc.
Qining Zhang: PhD student@University of Michigan (with Prof. Lei Ying)
Jinheng Zhang: PhD student@University of Pennsylvania (with Prof. Shirin Saeedi Bidokhti)
Yiqin Tan: master student@IIIS, Tsinghua University (with Prof. Longbo Huang). Now doing startup.

Academic Service

Reviewer

IEEE Journal on Selected Areas in Information Theory
IEEE Transactions on Information Theory

IEEE/ACM Transactions on Networking
IEEE Journal on Selected Areas in Communications
IEEE/ACM Transactions on Mobile Computing
IEEE Transactions on Communications
IEEE Transactions on Wireless Communications
IEEE Communication Letters
IEEE Wireless Communication Letters
Journal Communication Networks
NeurIPS 2024
AISTATS 2025
NeurIPS 2023 Workshop
IEEE International Conference on Computer Communications (Infocom) 2022, 2023
IEEE International Conference on Communications (ICC) 2022
IEEE Global Communications Conference (Globecom) 2021-2022
International Symposium on Information Theory (ISIT) 2020-2022
Information Theory Workshop 2022